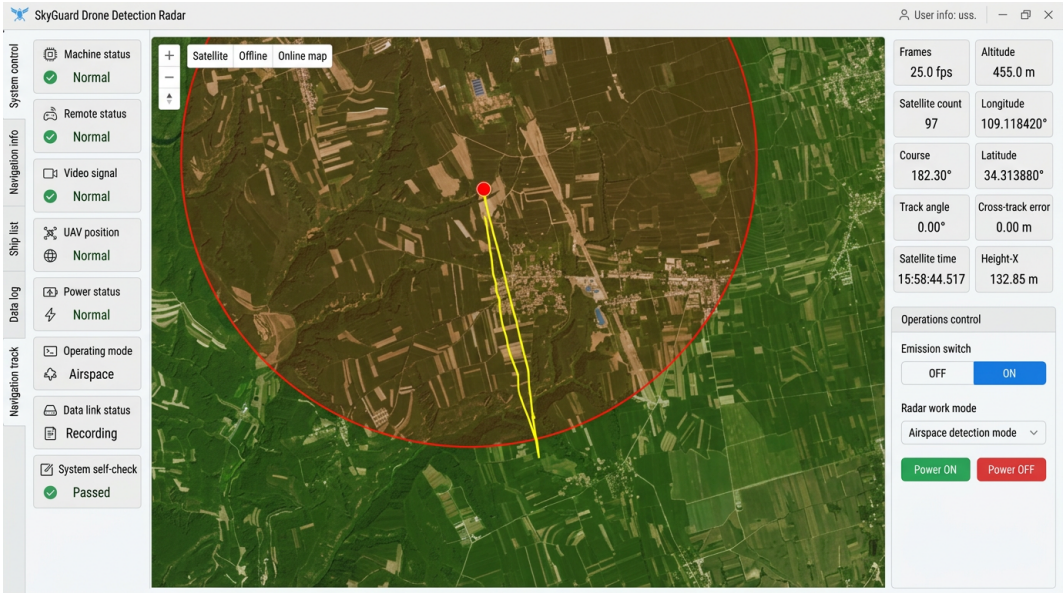
Sky-Guard Console

Digital Intelligent Flight Decision Support System



Background

As the low-altitude economy accelerates and public interest in recreational aviation grows, flight camps — serving general aviation, paragliding, drone testing, powered hang-gliders, hot-air balloons, and other light aircraft — are transitioning from experience-based management to data-driven, intelligent operations. Weather conditions such as wind speed, temperature, humidity, barometric pressure, cloud ceiling, and precipitation have a decisive impact on the safety and performance of light aircraft. The Sky-Guard Console bridges the gap between raw meteorological data and actionable flight decisions.

Core Capabilities

- Real-Time Weather Assessment — continuously evaluates whether conditions meet safe takeoff and landing thresholds for each aircraft type
- Flight Simulation — models flight attitude, climb rate, glide ratio, and control difficulty under varying meteorological scenarios
- Optimal Flight Planning — recommends best takeoff/landing direction, timing, climb profiles, holding patterns, and return-to-home waypoints
- Ideal Flight Window Prediction — forecasts time-slot-based operating windows to optimise camp scheduling and customer booking
- Automated Risk Alerts — generates flight-risk warnings and operational recommendations to support instructors and pilots

• Personalised Flight Advice — tailors recommendations based on site terrain, user profile, and spatiotemporal context

Personalised Camp Features

• Individual flight decision recommendations per visitor
• 3-day advance safety risk forecast
• Optimal flight period suggestions
• Visual flight-path preview and simulation
• Locally-tuned AI decision model that improves over time
• Emergency / sudden-change weather alerts

Solution Architecture

The system fuses real-time meteorological data ingestion, an aircraft performance parameter library, aerodynamic models, and AI-assisted decision algorithms into a unified platform. Data sources include gridded weather model outputs (1–2 km resolution), on-site sensor calibration, and historical operational records.

Service Tiers

Feature	Standard	Professional	Enterprise
Core Output	Index monitoring & review	Index-driven decision loop	Full custom pipeline
Forecast Horizon	24 h rolling	48 h rolling	72–120 h trend
Monitored Points	5 locations	10 locations	10+ (expandable)
AI Service	—	Multi-index analytics	Self-learning model
Alert Channel	Web dashboard	Real-time push	Multi-channel + audit trail
Deliverables	Flight data logging, 24 h forecast, 5-pt report, web portal	Daily report + video, 48 h forecast, 10-pt report, push notifications	Custom reports, behaviour logging, 72–120 h forecast, 10–15 pt report, multi-system push

Compliance & Risk Boundaries

• The service provides indices and advisory actions only; it does not replace any certified environmental data product.
• Indices serve as decision support, not the sole basis for safety or regulatory compliance. Operators must retain final authority and on-site verification procedures.
• When historical data is insufficient or significantly biased, model uncertainty will increase. The system flags such conditions and supports fallback to rule-based / conservative strategies.

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